```c
#include <stdio.h>

#define INF 9999

int main()
{
    int n, i, j, src, min, u;
    int cost[10][10], dist[10], visited[10];

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter cost adjacency matrix (9999 for no edge):\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &cost[i][j]);

    // Display vertices
    printf("\nVertices: ");
    for (i = 0; i < n; i++)
        printf("%d ", i);

    // Display adjacency matrix
    printf("\n\nAdjacency Matrix:\n");
    printf("    ");
    for (i = 0; i < n; i++)
        printf("%4d", i);
    printf("\n");

    for (i = 0; i < n; i++)
    {
        printf("%4d", i);
        for (j = 0; j < n; j++)
        {
            if (cost[i][j] == INF)
                printf("%4s", "INF");
            else
                printf("%4d", cost[i][j]);
        }
        printf("\n");
    }

    printf("\nEnter source vertex (0 to %d): ", n - 1);
    scanf("%d", &src);
```

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for (i = 0; i < n; i++)
{
    dist[i] = cost[src][i];
    visited[i] = 0;
}

dist[src] = 0;
visited[src] = 1;

for (i = 1; i < n; i++)
{
    min = INF;
    u = -1;

    for (j = 0; j < n; j++)
    {
        if (!visited[j] && dist[j] < min)
        {
            min = dist[j];
            u = j;
        }
    }

    if (u == -1)
        break;

    visited[u] = 1;

    for (j = 0; j < n; j++)
    {
        if (!visited[j] && cost[u][j] < INF &&
            dist[u] + cost[u][j] < dist[j])
        {
            dist[j] = dist[u] + cost[u][j];
        }
    }
}

printf("\nShortest distances from vertex %d:\n", src);
printf("-----------------------------\n");
printf("Vertex\tDistance\n");

for (i = 0; i < n; i++)
{
```

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        if (dist[i] >= INF)
            printf("%d\tUnreachable\n", i);
        else
            printf("%d\t%d\n", i, dist[i]);
    }

    return 0;
}
```